



GEOoptimize Inc
1-204-318-2156

220 Portage Avenue, Suite 1410
Winnipeg, Manitoba
Canada, R3C 0A5

Proposal for:
John Ciovacco
Aztech Geothermal
5 McCrea Hill Road, Suite 200
Ballson Spa, NY 12020

Proposal Date:
2022-02-28

Proposal Number:
QU-022

Understanding of project

Thank you for the opportunity to provide a proposal to perform the energy modelling and design of a ground heat exchanger (GHX) for the proposed Riverview Apartments at Cornith in the Town of Cornith, Saratoga County, NY.

The complex is being designed by Re4orm Architecture and will be owned by Rise Housing & Support Services. The proposed three-storey building will have 60 one, two and three-bedroom suites and be approximately 66,000 ft². The building will feature a distributed heat pump system for the suites, and a water-to-water heat pump will serve the domestic hot water system.

Proposed design methodology

Phase 1: Energy Model

The size, cost, and performance of a GHX is sensitive to the peak heat transfer to and from the ground. The GHX is also sensitive to the total heat rejection to the ground when the building is being cooled compared to the amount of heat extracted from the ground to heat the building, known as the annual energy balance.

An accurate hourly energy model of the building connected to a GHX is critical to the development of a GSHP system that is cost-effective to build, performs efficiently and predictably for the life of the building and is easily maintained. Energy loads are determined by simulating building performance based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data for the location.

Using the hourly energy model to determine the impact of changes to glass specifications, lighting and lighting controls, ventilation air strategy, domestic hot water production, distribution system design, etc. can help to balance the loads on the GHX, resulting in a reduction in the size and cost of the GHX and affects the efficiency and performance of the system. If the design team is receptive, GEOoptimize can develop several iterations of the energy model and the impact of the changes on the size, cost, and performance of the GHX.

The resultant hourly energy model provides the information needed to calculate the configuration, size, and cost of the GHX. Hourly energy loads also allow us to calculate the cost of heating, cooling, and producing domestic hot water for the facility and to estimate CO₂ emissions from the building.

Phase 2: Detailed GHX Design Calculations

The land area available for the construction of the GHX is determined with the building owners. Consideration is given to all potential GHX designs and configurations based on building energy loads, geology, land area available for construction of the GHX and on the capabilities of local GHX contracting companies. Potential GHX designs may include vertical borehole configurations, horizontally excavated or horizontally bored GHX fields as well as groundwater options.

Thermal properties of the soil and rock at the site are estimated with a review of available geological information from water well databases, interviews with drilling and excavation contractors and geologists familiar with the area.

GEOptimize will develop the detailed design of the GHX for the facility. The design will include:

- Layout of GHX circuit piping for the GHX,
- Configuration of fusion welded headers to connect GHX circuits to supply / return runout piping
- Supply / return runout piping from GHX headers to manifold in the building
- Specifications for GHX piping for circuits, headers, and supply / return runout piping
- Specifications for heat transfer fluid and GHX fluid volume
- GHX manifold design up to isolation valves between heat pump system piping and GHX
- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control

GEOptimize will work closely with Aztech Geothermal and the mechanical engineer of record for the interior mechanical system to ensure the proposed mechanical equipment and distribution system design is optimized to operate efficiently with the proposed GHX.

Information required

The following information is required to accurately model the heating and cooling loads of the building and to predict the performance of the GHX:

- Architectural, mechanical, and electrical drawings for the project
- Window specifications for the project including U-value and solar heat gain coefficient (SHGC)
- Site plan indicating underground services and areas available for construction of GHX
- Estimated occupancy of the facility
- Gas and electricity detailed rate structure including time of use rates and demand penalties (if applicable)

Scope of work

Phase 1: Feasibility of GSHP system

GEOptimize scope of work includes:

- Detailed iterative hourly energy model working in collaboration with architectural and engineering design team. The energy model will be completed in Trane Trace 700. Input assumptions and results of the energy model results will be shared with Aztech Geothermal and the architectural and engineering design team to provide equipment design loads.

Phase 2: Detailed GHX design

Details of the ground heat exchanger (GHX) provided will include:

- Layout of GHX circuit piping for the recommended GHX configuration
- Design of fusion welded headers to connect GHX circuits to supply / return runout piping
- Supply / return runout piping from GHX headers to manifold in the building
- Specifications for heat transfer fluid and estimated heat transfer fluid volume
- GHX manifold design up to isolation valves between heat pump system piping and GHX
- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control

Pricing

Description	Rate	Qty	Line Total
Phase 1: Energy Model (as outlined in Scope of Work) Iterative hourly energy modelling of the proposed building based on information and drawings provided	\$13,900.00	1	\$13,900.00
Phase 2: Detailed GHX design (as outlined in Scope of Work)	\$5,500.00	1	\$5,500.00
Proposal Total (USD)			\$19,400.00

Terms

Additional work out of scope will be charged at \$200/hr. Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

Authorized signature – Aztech Geothermal